

Data Cleaning Rules

This document describes the data cleaning logic used in the online school performance analytics project. The rules were defined after the data dictionary preparation and the initial data audit.

1. General Cleaning Rules

- Raw files from the data/raw folder are not modified.
- All transformations are performed in Python.
- Cleaned datasets are saved to the data/processed folder.
- Before deleting, changing or excluding data from calculations, the processing reason is documented.
- CRM identifiers (Id, CONTACTID, Contact Name) are processed as string values.
- Identifiers are not used as numeric metrics.
- If long identifiers are read as float values, precision may be lost; therefore, cleaned string versions of identifiers are used for relationships between tables.
- Technical fields created for checks and cleaning are added only to working dataframe copies.
- Rows are not removed automatically if they can still be preserved for overall analysis and restricted only in specific calculations.
- If a row is not suitable for a specific analysis, it is excluded only from that analysis and may remain in the general dataset.

2. Spend Cleaning Rules

The Spend table is used to analyze marketing spend, impressions, clicks and advertising source performance.

- Date, Source, Impressions, Spend and Clicks are key fields for marketing analysis.
- Date is converted to datetime format.
- Impressions, Spend and Clicks are converted to numeric format.
- According to the audit results, there are no missing values in Date, Source, Impressions, Spend and Clicks, so no additional missing-value processing is required for these fields.
- Negative values in Impressions, Spend and Clicks are treated as errors and are not used in calculations.

Source

- Source is used as the main advertising channel.
- Source values should belong to the list of valid sources.
- If a value is written in a different format but the source is clear, it is standardized to a single naming convention.
- If the source is unknown or cannot be interpreted, it is marked as Unknown or excluded from source-level analysis.

Campaign

- Campaign is used to analyze advertising campaign performance.
- If Campaign is missing and Spend = 0, the value remains missing.
- If Campaign is missing and Spend > 0, a technical campaign value is created using the Date_Source rule.
- Dates inside campaign names are not used as the actual campaign launch date.
- If different spellings of the same campaign are found, values are standardized where possible without manual guessing.

AdGroup and Ad

- Missing values in AdGroup and Ad are not filled because they may indicate that the data is not detailed down to the ad group or ad level.
- AdGroup and Ad are used only for additional analysis.
- If a value cannot be interpreted, it may be marked as Unknown.
- If an advertising field cannot be matched with the corresponding field in Deals, the row is not removed, but it is not used for detail analysis at that level.

Full Duplicates

- Full duplicates are not removed automatically.
- According to the audit results, duplicates contain almost no advertising activity and do not materially affect Spend, Impressions or Clicks.
- When analyzing row counts and record distribution by source, full duplicates are considered separately.
- If further analysis is based on the number of records, duplicates should either be excluded or their impact should be explicitly stated.

Clicks > Impressions

- For rows where Clicks > Impressions, the invalid_ctr flag is created.
- These rows are not removed because they still contain spend and click information.
- Rows with Clicks > Impressions can be used for:
 - spend analysis;
 - click analysis;
 - CPC calculation;
 - linking advertising data with leads and deals.
- Rows with Clicks > Impressions are not used for CTR calculation or impression efficiency analysis.

3. Contacts Cleaning Rules

The Contacts table is used as a CRM contact reference table and to connect Deals and Calls.

Id

- Id is converted to string format.
- Id is used as a technical contact identifier.

Contact Owner Name

- Contact Owner Name is used as a categorical field.
- If different spellings of the same manager are found, values are standardized.
- If the value is missing or cannot be interpreted, it is filled as Unknown.
- Rows with Unknown are not removed, but they may be excluded from manager-level analysis.

Created Time and Modified Time

- Created Time and Modified Time are converted to datetime format.
- According to the audit results, the Contacts table does not require significant additional cleaning.

4. Calls Cleaning Rules

The Calls table is used to analyze sales team calling activity and to link calls with contacts.

Id and CONTACTID

- Id and CONTACTID are converted to string format.
- These fields are technical CRM identifiers, not numeric metrics.
- The cleaned CONTACTID field is used to link Calls with Contacts.
- After processing CONTACTID as a string, all filled CONTACTID values from Calls are correctly matched with Contacts.Id.
- Rows without CONTACTID remain in the general call activity analysis, but are not used for relationships with Contacts and Deals.

Call Start Time

- Call Start Time is converted to datetime format.
- Rows with invalid Call Start Time are not used in call time-series analysis.

Call Duration (in seconds)

- Call Duration (in seconds) is converted to numeric format.
- Rows with missing or invalid Call Duration remain in the table, but are not used in call duration analysis.
- Negative values in Call Duration (in seconds) are treated as errors and are not used in duration analysis.
- Missing call duration values are not automatically filled with zero, because a missing value is not the same as a short call.

Call Owner Name, Call Type, Call Status

- Call Owner Name, Call Type and Call Status are used as categorical fields.
- If different spellings of the same value are found, they are standardized.
- If a value is missing or cannot be interpreted, it is filled as Unknown.
- Rows with Unknown remain in the general analysis, but may be excluded from detail analysis by the relevant field.

Dialled Number and Tag

- Dialled Number and Tag are excluded from further analysis because, according to the audit, they are fully empty.
- These fields are not restored.

Outgoing Call Status and Scheduled in CRM

- Outgoing Call Status and Scheduled in CRM remain as additional fields.
- They are used only when a separate analysis of outbound or scheduled calls is required.
- Missing values in these fields are allowed and are not filled automatically.

5. Deals Cleaning Rules

The Deals table is the main table of the project. It is used to analyze the funnel, payments, products, sources, managers and revenue.

Key Fields

- Id is converted to string format.
- Id is used as a technical deal identifier.
- Rows without Id, Stage, Source or Created Time are excluded from deal analysis because they cannot be used correctly for funnel, source and deal creation dynamics analysis.

Stage

- Stage is the main field used to determine the current deal status.
- Stage values are standardized if different spellings are found.
- Rows without Stage are excluded from funnel and conversion analysis.
- The main paid deal condition is Stage == Payment Done.
- Payment Done means that the money has been received.
- Other statuses are used for funnel analysis, but are not treated as payment confirmation.

Deal Owner Name

- Deal Owner Name is used to analyze sales manager performance.
- If different spellings of the same manager are found, values are standardized.
- If the manager is missing, the row remains in the general analysis but is not used in manager-level analysis.
- All managers from Deals.Deal Owner Name are found in Calls.Call Owner Name, so the manager reference is considered consistent.
- Deal Owner Name and Call Owner Name are used for aggregated manager-level analysis, but not as a strict key for linking specific deals and calls.

Created Time and Closing Date

- Created Time is converted to datetime format.
- Closing Date is converted to date format.
- Created Time contains date and time, while Closing Date contains only date.
- For logical date checks and deal duration calculation, calendar dates without time are used.

- If Closing Date is earlier than Created Time, the row is not removed automatically.
- For such rows, the invalid_deal_dates flag is created.
- Rows with invalid_deal_dates = True remain in general analysis of deals, statuses, sources and payments.
- Rows with invalid_deal_dates = True are not used to calculate deal duration or analyze time from creation to closing.

SLA

- SLA is converted to time / timedelta format.
- Rows with missing SLA remain in the table.
- Rows with missing or invalid SLA are not used in lead response speed analysis.
- Negative or non-interpretable SLA values are marked as errors and excluded from SLA analysis.

Quality

- Quality is used to analyze lead quality.
- Values are standardized to the following categories:
 - A - High
 - B - Medium
 - C - Low
 - D - Non Target
 - E - Non Qualified
 - F
- Unknown or invalid values are converted to Unknown.
- Missing values are filled as Unknown to preserve rows in the analysis and separately track deals without quality assessment.

Lost Reason

- Lost Reason is not filled automatically.
- Missing values in Lost Reason remain as they are.
- Lost Reason is not used as an independent indicator of a lost deal.
- Stage is the main field used to determine deal status.
- Lost Reason analysis is performed only together with Stage.
- If Lost Reason is filled when Stage is not Lost, the value is not removed: it may reflect historical information or CRM usage specifics.
- Duplicate as a lost reason may be used for additional lead duplicate checks, but it is not removed automatically without verification.

Product and Education Type

- Product and Education Type are critical only for deals with Stage = Payment Done.
- Missing values in Product and Education Type for deals with statuses other than Payment Done are not filled and are not treated as errors.
- If Product is missing for a Payment Done deal, it is filled as Unknown.
- If Education Type is missing for a Payment Done deal, it is filled as Unknown.
- Payment Done deals with Product = Unknown or Education Type = Unknown remain in payment, revenue and conversion analysis.
- In product analysis, such deals are shown as a separate Unknown category to avoid distorting the distribution of known products and education formats.
- If different spellings of the same product or education format are found, values are standardized.

Payment Type

- Payment Type is used to analyze payment methods.
- Missing Payment Type values are allowed for unpaid deals.
- For Payment Done deals, missing Payment Type is filled as Unknown.
- In payment type analysis, deals with Payment Type = Unknown are shown as a separate category.
- If different spellings of the same payment type are found, values are standardized.

Financial Fields and Payment Logic

Financial fields

- Initial Amount Paid
- Offer Total Amount

Technical cleaning of amounts

- Initial Amount Paid and Offer Total Amount are converted to a consistent numeric format.
- Before conversion, the following elements are removed:
 - currency symbols (€, \$);
 - spaces;
 - thousand separators;
 - extra technical symbols.
- Thousand and decimal separators are converted to a consistent format.
- Euro and dollar values are not recalculated into different currencies because, within this project, amounts are used as CRM values from the export.
- The currency symbol is removed only for technical field normalization.
- If a value cannot be converted to a number after cleaning, the row is not used in financial calculations.

Payment confirmation

- The main paid deal condition is Stage == Payment Done.
- Initial Amount Paid is used as the amount of the actually received first payment only for deals with Stage = Payment Done.
- If Stage == Payment Done but Initial Amount Paid is missing, the deal is included in payment conversion but excluded from revenue and average check calculations.
- If Initial Amount Paid > 0 but Stage != Payment Done, the amount is not treated as actual revenue.
- Offer Total Amount is used as the offer amount / potential deal value, not as payment confirmation.
- Low values in financial fields, such as 0, 1 and 9, are not removed automatically because they may reflect demo access or symbolic payments.
- For average check analysis, an additional metric excluding symbolic payments may be calculated, but the original values are preserved.

Financial field logic

- If Initial Amount Paid > Offer Total Amount, the row is not removed automatically.
- For such rows, the initial_amount_gt_offer flag is created.
- Such cases may be related to recalculations, changes in offer value, refunds or CRM payment recording specifics.
- Because the dataset does not include detailed treasury transactions or refund data, amounts are not adjusted manually.
- Rows with initial_amount_gt_offer = True remain in overall payment and conversion analysis.
- When analyzing the relationship between the first payment and the total offer amount, such rows are shown separately.

Course duration and Months of study

- Course duration and Months of study are used only for education and student analysis.
- Values are converted to numeric format.
- Negative values are treated as errors and are not used in education duration analysis.
- If Months of study > Course duration, the row is flagged as a logical error.
- Missing values are allowed if the client did not become a student or education did not start.
- According to the FAQ, if Months of study is missing, the student is not studying.
- Rows without Months of study are not used in education and retention analysis.
- When calculating revenue, payment is determined by Stage == Payment Done, not by Months of study.

City

- City is used for geographic analysis.

- Missing values, the value -, technical noise and non-interpretable values are converted to Unknown.
- Different spellings of the same city are standardized where possible without manual guessing.
- Rows with City = Unknown remain in general analysis of deals, payments and sources.
- For geographic analysis, rows with City = Unknown are either excluded or shown as a separate Unknown category.

Level of Deutsch

- Level of Deutsch is used to analyze the relationship between German language level and deal success.
- Language level values are standardized to:
 - A1
 - A2
 - B1
 - B2
 - C1
 - C2
- Different spellings of the same level are standardized.
- Free-text values, invalid levels and non-interpretable values are converted to Unknown.
- Missing values are converted to Unknown if the field is used in visualizations and groupings.
- For analysis of the relationship between German language level and performance, only filled/standardized values are used; Unknown is shown separately.

Page

- Page is used as an additional field for landing page analysis.
- Missing values are not critical.
- If the value is missing, the row remains in the general analysis.
- If needed, missing values may be marked as Unknown or No page.
- Different spellings of the same page are standardized if required for analysis.

Advertising Fields and the Deals-Spend Relationship

Advertising fields in Deals

- Source
- Campaign
- Term
- Content

Advertising fields in Spend

- Source
- Campaign
- AdGroup
- Ad

Source

- The main relationship between Deals and Spend for marketing analysis is built on Source.
- Source is considered the most reliable matching level because all sources from Deals are found in Spend.
- Missing Source values in Deals are not allowed for source-level analysis.
- If Source cannot be restored, the row is excluded from channel-level analysis but may remain in the general deal analysis if other key fields are filled.

Campaign

- Campaign is used as additional detail.
- Because campaigns only partially match between Deals and Spend, Campaign is not used as the only key for joining data.
- Missing Campaign values are not filled automatically if the lead did not come from an advertising campaign.
- Rows with missing Campaign are not used for campaign analysis.

- If Campaign does not match between tables, the row is not removed but is not used for campaign-level detail analysis.

Term / AdGroup

- Deals.Term may correspond to Spend.AdGroup.
- Deals.Term and Spend.AdGroup are used only as supporting fields because the match rate between them is low.
- Missing values are not filled automatically.
- Rows without Term or without a matching AdGroup are not used for ad group-level detail analysis.

Content / Ad

- Deals.Content may correspond to Spend.Ad.
- Deals.Content and Spend.Ad may be used for additional ad/creative analysis, but only for matching values.
- Missing values are not filled automatically.
- If a value cannot be matched between tables, the row remains in general analysis but is not used for ad-level detail analysis.

Deals and Contacts Relationship

- Contact Name in the Deals table is processed as a CRM contact identifier, not as a customer name.
- The cleaned string value of Contact Name is used to link Deals with Contacts.
- Deals without Contact Name remain in the general deal analysis, but are not used for joins with Contacts.
- If Contact Name is not found in Contacts, the row is not removed automatically but is excluded from relationship analysis between tables.
- According to the audit results, the Deals.Contact Name → Contacts.Id relationship is considered reliable enough: 21,531 of 21,532 filled values were matched.

6. Final Logic for Using Cleaned Data

- Deals is the main table for analysis.
- Payment confirmation is determined by Stage.
- The main paid deal status is Payment Done.
- Actual revenue is calculated from Initial Amount Paid only for deals with Stage = Payment Done.
- Payment Done deals without payment amount are included in conversion, but excluded from revenue and average check calculations.
- Spend is used to analyze marketing spend.
- The main relationship between marketing spend and deals is built on Source.
- Campaign, Term, Content, AdGroup and Ad are used only as additional detail.
- Calls is used to analyze sales activity and can be linked to Contacts via CONTACTID.
- Contacts is used as a contact reference table.
- All relationships between tables are built using cleaned string identifiers.
- Rows with data quality flags are not removed automatically, but are excluded from calculations where the identified issue is critical.